

Pick a High-Value Item with Dual Scan SOP

One episode picks one high-value item and seals it. It starts with one item in the pick bin, the location card standing in its clip, an open tote with its lid up, one tamper tag lying flat, and a log card with an open box. It ends with the item shut inside the tote, the tamper tag pressed across the lid seam, the seal stub pressed in the next open box on the log card, and every card back on its spot. Twelve episodes fill one log card. One log card is one session.

The order never changes: scan the location, pick and scan the item, drop it in the tote, scan the witness card, close and seal the tote, log the seal number. Nothing goes in the tote until both the location and the item have read good. The tote is never closed until the witness card has read good, and the seal never goes on until the lid is shut.

The gripper does all the work. It shows the location card, the item and the witness card to the scanner, it puts the item in the tote, it closes the lid, it presses the seal, and it presses the stub. The scanner never moves and the tote never leaves its spot. The tote is the only thing that ever holds the item, and it stays where it is.

Only one thing is held at a time. Nothing is passed from one grip to another, no grip is changed while an object is off the table, and nothing is dragged or slid along the table. Every pick lifts straight up off the surface before it moves anywhere.

Every item in a session comes from the same pick bin and goes in its own tote, one item per tote. The counts are fixed: one item, three good reads, one tag, one stub and one box on the log card per episode, one press of 2 seconds on each sticky part, and up to three tries per scan. There is no second item in a tote, no second tag on a tote, and no second stub in a box. Anything that lands on the table is left exactly where it falls, tagged, and the episode goes on. Nothing is ever tidied up on camera.

Setup

Finish both checklists before starting an episode.

Cell configuration

Environment camera: 90 mm.

cell_type: single_arm

Hardware checklist

1. Cameras are on and recording.

2. The environment camera frame holds the whole tabletop in one shot: the pick bin, the location clip, the tote spot, the scanner, the card spot, the tag spot, the log spot and the reject spot.
3. The item stays in frame from the moment it leaves the bin until the lid is shut, and the scanner window stays in frame for every scan.
4. The seal number on the tag and the number on the stub are both readable in the frame before the tag is picked up.
5. The gripper is at home and open.
6. The tabletop is dry and holds nothing but the items in the Materials checklist.
7. The scanner is on. It gives one beep and a green light on a test read of the spare test card.
8. The gripper reaches the pick bin, the location clip, the scanner window, the tote, the card spot, the tag spot and every box on the log card without stretching to a joint limit or folding in on itself. Bring a spot forward until this is true.
9. The paths from the bin to the scanner, from the scanner to the tote, from the card spot to the scanner, and from the tag spot to the tote are clear. Carry one item along each path by hand during setup: it must clear the tote walls and the standing lid. Widen the gaps until it does.

Materials checklist

1. Twelve high-value items stand in the pick bin at the back left, one taken per episode. They are the same size and shape, so one grip works on all twelve.
2. Each item carries one item label on one face. The label is flat, dry and unbroken, and its code is clean enough for the scanner to read.
3. Each item is square enough for the gripper to close on it squarely and hold it level, and small enough to lie flat in the tote with room to spare.
4. One location card stands upright in a clip on the front of the pick bin, with its location label facing out. The card is stiff, so it does not bend when the gripper closes on it.
5. The clip holds the card and lets it go without a pull. A clip that grips so hard the bin lifts with the card is replaced.
6. One witness card lies flat on the card spot at the front left, printed side up. It is the same card in every episode.
7. One open tote stands at the tote spot in the middle of the table, empty and dry, with its lid up and leaning back so the mouth is clear.
8. The tote is deep enough to hold the item below the rim and shallow enough that the item goes straight down into it without being pushed.
9. The tote grips the table. Push it by hand during setup: if it slides or lifts, weigh it down or clamp it. Nothing pins the tote during the episode.
10. The lid falls shut under its own weight once it is past upright. A lid that has to be forced or held down is replaced.

11. One tamper tag lies flat on the tag spot at the front middle, sticky side down on its backing, with its stub end toward the front. Its corner is lifted off the backing, so the gripper has an edge to close on.
12. The seal number printed on the tag body and the number printed on the stub are the same, and no two tags in a session carry the same number.
13. One log card lies flat on the log spot at the back right, with twelve numbered boxes, three across and four deep, and at least one box open.
14. The reject spot at the front right is empty. It takes items and tags that fail and nothing else.
15. A clear gap of at least a gripper's width sits around the tote on every side, so the item goes straight down into it without touching a wall or the standing lid.

Workspace layout

Three rows: the pick bin and the log card at the back, the tote in the middle with the scanner beside it, the witness card and the tamper tag at the front. Each item travels forward to the scanner, back to the tote, and stays there.

Back row, left to right:

1. **Pick spot**, back left. The pick bin stands here with the items in it and the location card standing in its clip on the front of the bin. It empties as the log card fills.
2. **Log spot**, back right. The log card lies flat here with its twelve boxes. It fills as the pick bin empties.

Middle row, left to right:

1. **Tote spot**, the middle of the table. The tote stands here with its lid up. Everything that happens to the item after the scan happens here.
2. **Scan spot**, at the right edge, the scanner standing with its window facing the tote.

Front row, left to right:

1. **Card spot**, front left. The witness card lies flat here, printed side up.
2. **Tag spot**, front middle. One tamper tag lies flat here on its backing with its corner lifted.
3. **Reject spot**, front right. It takes an item that will not read and a tag that comes away bad.

No carry passes over the open tote on the way to somewhere else, no carry passes over the log card, and nothing travels to the far side of the table.

Vocabulary

Item: one high-value thing that is picked, scanned and sealed. One item per episode.

Item label: the code printed on one face of the item. Every item has one.

Pick bin: the open bin at the back left that the items are taken from.

Location card: the stiff card standing in the clip on the front of the pick bin. It carries the location label and it never goes anywhere but the scanner and back.

Location label: the code printed on the location card. It says where the item was picked from.

Dual scan: one good read of the location label and one good read of the item label, in that order, for the same item, both before the item goes in the tote.

Witness card: the printed card at the card spot that is scanned to witness the pick. It is shown to the scanner while the item lies in the open tote.

Scanner: the fixed reader at the right edge. Its window is its reading face.

Scan gap: the 5 to 10 cm of daylight between the code and the scanner window, held for the whole scan try.

Scan try: one hold of a code in front of the window, square to it, for up to 3 seconds. Three tries is the limit for each code.

Good read: one beep and a green light from the scanner. Nothing else counts as a read.

Tote: the box at the tote spot that the item is sealed in. It holds one item.

Lid: the hinged top of the tote. It stands up and leaning back at the start and lies shut at the end.

Lid seam: the line where the shut lid meets the front wall of the tote. The seal goes across it.

Shut lid: the lid lying down flat on the tote with no gap along the seam and nothing caught under it.

Tamper tag: the numbered sticker that seals the tote. It has two parts: the seal body and the seal stub.

Seal body: the larger part of the tag. It goes across the lid seam, half on the lid and half on the front wall.

Seal stub: the small part of the tag that peels off. It carries the same number as the seal body and it goes on the log card.

Seal number: the number printed on both parts of the tag. It is the number that gets logged.

Sealed tote: the lid shut and the seal body pressed flat across the seam, with no lifted corner and no gap into the tote.

Press: the closed gripper pushed onto a sticky part for 2 seconds, from the middle out to each of its edges.

Log card: the printed card at the log spot with twelve numbered boxes. One stub goes in one box.

Next open box: the first empty box in the fill order, the front row left to right, then the row behind it, ending at the back right box as box twelve.

Full log card: twelve boxes, one stub in every box.

Bad tag: a tag that is torn, folded onto itself, doubled with a second tag, or stuck down somewhere it should not be. It goes to the reject spot.

Reject item: an item that will not read after three scan tries. It goes to the reject spot.

Hold: a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.

Stable placement: the object stands still after release, flat on its surface, and does not rock, tip, slide or lean on anything.

Spill: an item, a card, a tag or a stub that ends up outside the bin, the tote, a clip, a spot or a box. It is left where it falls and cleared in the reset, never on camera.

Home: the resting pose, clear of the table with the gripper open.

Steps

Step 1: Scan the location

Goal: one good read of the location label, and the location card standing back in its clip.

The location is read first, so every item that follows is tied to the place it came from. Nothing is picked out of the bin before this read is done.

1.1 Take the location card and show it

Close on the top edge of the location card and lift it straight up out of its clip.

Carry it to the scanner and hold the location label in front of the window at the scan gap, square to the window, and hold it still for up to 3 seconds.

If the scanner beeps once and shows green, go to 1.2.

If nothing reads within 3 seconds, move the card 2 to 3 cm closer to the window or farther from it and try once more. **Three scan tries in total, and no more.**

Check: the scanner has given one beep and a green light for the location. If there is no good read after three tries, put the card back in its clip, end the episode, and start a new one.

1.2 Put the location card back

Carry the card back to the clip, lower it in until it stands on its own, then open the gripper and lift clear.

Check: the card stands upright in its clip with the location label facing out, and the pick bin has not moved.

Step 2: Pick the item and scan it

Goal: one good read of the item label on the item that is in the gripper.

The item stays in the gripper for this whole step. It is not set down between the pick and the read.

2.1 Pick one item

Close on one item in the pick bin, square to it, so it hangs level.

Lift it straight up out of the bin before any sideways move.

Take one item per episode. Never lift two at once, and never drag or slide an item across the table.

Check: one item hangs level in the gripper and nothing else in the bin has moved. If two came away together, set both back in the bin and take one again. Make no more than two extra tries.

2.2 Show the item label

Carry the item to the scanner and hold the item label in front of the window at the scan gap, square to the window, and hold it still for up to 3 seconds.

If the scanner beeps once and shows green, go to Step 3.

If nothing reads within 3 seconds, turn the item until the label faces the window, or move it 2 to 3 cm closer to the window or farther from it, and try once more. **Three scan tries in total, and no more.**

Only the item label is shown to the scanner. Never show another face of the item and never show a second item.

Check: the scanner has given one beep and a green light for this item, and this is the second good read of the episode. If there is no good read after three tries, put the item at the reject spot, end the episode, and start a new one. The item does not go in the tote.

Step 3: Put the item in the tote

Goal: the item lies flat in the open tote, below the rim, with the lid still up.

Carry the item from the scanner to the tote, level the whole way, and bring it over the mouth of the tote.

Lower it straight down into the tote until it is resting on the bottom, then open the gripper and lift clear. Open the gripper low, no more than 2 cm above the item, and never drop the item in.

One item per tote. Never put a second item in a tote that already holds one.

Check: one item lies flat in the tote, below the rim, not leaning on a wall and not caught under the lid. The lid still stands up and leaning back. If the item leans or sits high, lift it clear and place it again. Make no more than one correction.

Step 4: Scan the witness card

Goal: one good read of the witness card while the item lies in the open tote, and the card back on its spot.

The tote stays open for this whole step, so the camera sees the item and the witness read together. The lid does not move until this read is done.

4.1 Take the card and show it

Close on the edge of the witness card at the card spot and lift it straight up off the table.

Carry it to the scanner and hold its printed side in front of the window at the scan gap, square to the window, and hold it still for up to 3 seconds.

If the scanner beeps once and shows green, go to 4.2.

If nothing reads within 3 seconds, move the card 2 to 3 cm closer to the window or farther from it and try once more. **Three scan tries in total, and no more.**

Check: the scanner has given one beep and a green light for the witness card, and this is the third good read of the episode. If there is no good read after three tries, lay the card back on its spot, take the item out of the tote and put it at the reject spot, end the episode, and start a new one.

4.2 Put the witness card back

Carry the card back to the card spot, lay it flat printed side up, open the gripper and lift clear.

Check: the card lies fully on its spot, printed side up, not draped over the tote or the table edge.

Step 5: Close and seal the tote

Goal: the lid shut and the tamper tag pressed flat across the lid seam, with the seal number in view.

The tote stays on the table for this whole step and is never lifted.

5.1 Close the lid

With the gripper closed and empty, push the standing lid forward until it falls past upright and lies down on the tote.

Press the lid down along the seam until it lies flat, then lift clear.

Check: the lid lies flat with no gap along the seam and nothing caught under it. If the lid stands part way or an edge is caught, lift the lid, clear it, and close it again. Make no more than one correction.

5.2 Peel the tag and place it

Close on the lifted corner of the tamper tag and peel it off its backing in one move, keeping the printed side up the whole way.

Carry the tag to the tote and hold it 1 to 2 cm above the lid seam, still clear of it.

Turn it until it runs across the seam at a right angle, then move it until half of the seal body sits on the lid and half sits on the front wall, with the seal number facing the camera.

Make no more than two small moves to line it up. **The tag is lined up before it touches, not after.**

Lower it straight down onto the seam until the whole sticky side is touching, then open the gripper and lift clear.

Check: one whole tag lies across the seam with the seal number in view. If the tag tears, folds onto itself, or a second tag comes away with it, lay it at the reject spot, open the gripper, and go on with the next tag once. Make no more than one extra peel.

5.3 Press the seal flat

Close the gripper and press the seal body down for 2 seconds, starting at the middle and working out to each of its edges.

Bring the gripper straight up off the seal at the end of the press.

Once the tag is down it is not slid, nudged or turned. A tag that landed crooked is left where it is and the episode is tagged for review.

Check: the seal body lies flat across the seam with no lifted corner and no crease, half on the lid and half on the front wall, and no gap into the tote. If one corner lifts, press that corner down once more. Make no more than one correction.

Step 6: Log the seal number

Goal: the seal stub pressed in the next open box on the log card, with its number in view, and the fill order unbroken.

6.1 Peel the stub

Close on the stub end of the tag and peel the stub off the seal body in one move, keeping its printed side up.

Lift it clear of the tote and carry it to the log card. The seal body stays on the tote and does not lift with the stub.

Check: the stub hangs from the gripper printed side up, and the seal body is still flat across the seam with no lifted corner. If the seal body lifts with the stub, press it back down once, then go on.

6.2 Press the stub in the next open box

Bring the stub over the next open box and hold it 1 to 2 cm above the card, still clear of it.

Fill the boxes in order: the front row left to right first, then the row behind it, ending at the back right box as box twelve. Never skip a box and never press a second stub into a box that already holds one.

Lower the stub straight down inside the lines of the box, then open the gripper and lift clear. Close the gripper and press the stub down for 2 seconds, from the middle out to each of its edges.

Check: one stub lies inside the lines of the right box, its number is in view from above, and no corner lifts. If a corner lifts, press it down once more. If the stub sits outside the lines of the box or over two boxes, leave it where it is and let review flag the episode.

Step 7: End the episode

Check the tote: shut, sealed across the seam, standing in its spot, holding one item.

Check the numbers: the number on the seal body and the number on the stub are the same and both are in view.

Check the log card: the stub in the next open box in the fill order, flat and inside the lines.

Check the cards: the location card upright in its clip, the witness card flat on its spot.

Fix a failed check only with the fix written in its step. If no fix is written, or the fix would need an earlier step run again, end the episode and reset instead.

Go home with the gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the sealed tote away for the next stage and stand a fresh empty tote at the tote spot, dry inside, with its lid up and leaning back.
2. Lay one fresh tamper tag flat on the tag spot, sticky side down on its backing, stub end toward the front, with its corner lifted off the backing. Check that the number on the body and the number on the stub match, and that the number has not been used in this session.
3. Leave a full log card where it is, take it away for filing, and lay a fresh log card flat on the log spot with all twelve boxes open.
4. Empty the reject spot. Set any reject item or bad tag aside for a person to look at, and do not put either back in the flow.
5. Top up the pick bin so it holds enough items to finish the log card, each with one clean item label on one face.
6. Stand the location card upright in its clip with the location label facing out, and lay the witness card flat on the card spot printed side up.
7. Wipe any loose backing paper, stub or dust off the tabletop, out of the tote and off the log card, and dry them.
8. Check that the tote still grips the table and its lid still falls shut on its own, the clip still holds the location card and lets it go, the scanner still reads the spare test card, and no item label or card is smeared or torn.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

Visible cue: the item leaves the bin before the location card has read good, the item goes in the tote before its own label has read good, the lid is closed before the witness card has read good, the tag goes on before the lid is shut, the stub is pressed before the seal body is pressed, or a step is skipped.

SOP rule broken: Steps 1 to 6, scan the location, pick and scan the item, tote it, witness it, seal it, log it, in that order.

Coaching note: location first, then the item, then the witness, then the seal, then the log. Nothing goes in the tote on one read alone.

Violation: Dual scan not done

Visible cue: only one of the two codes is shown to the scanner before the item goes in the tote, the same code is read twice and counted as two, or the item is toted after a red light on either code.

SOP rule broken: Steps 1.1 and 2.2, one good read of the location label and one good read of the item label, both before the item goes in the tote.

Coaching note: two different codes, two beeps, two greens, both before the item drops. One read is not a dual scan.

Violation: Witness scan skipped or done at the wrong time

Visible cue: the lid is closed with no witness read, the witness card is shown before the item is in the tote, or it is shown after the lid is already shut.

SOP rule broken: Step 4, the witness card is read while the item lies in the open tote and before the lid moves.

Coaching note: the camera has to see the item in the open tote and the witness read at the same time. Card first, lid second.

Violation: Wrong code shown

Visible cue: another face of the item, another card, another item, or the seal number is held in front of the scanner window in place of the code the step asks for.

SOP rule broken: Steps 1.1, 2.2 and 4.1, each step shows its own code and nothing else.

Coaching note: show the code the step names. A read of the wrong code is not a read.

Violation: More than three scan tries

Visible cue: a code is held in front of the window a fourth time instead of stopping at three.

SOP rule broken: Steps 1.1, 2.2 and 4.1, three scan tries for each code, and no more.

Coaching note: the count is fixed at three. After the third, the episode ends where its step says.

Violation: More than one item picked or toted

Visible cue: two items come away from the bin and both are carried on, or a second item is lowered into a tote that already holds one.

SOP rule broken: Steps 2.1 and 3, one item per episode and one item per tote.

Coaching note: one item, one tote, one seal, one box. If two come away, both go back and one is taken again.

Violation: Item dropped into the tote

Visible cue: the gripper opens more than 2 cm above the item, the item falls the last part of the way, or it lands leaning on a wall or caught under the lid.

SOP rule broken: Step 3, lower the item into the tote and open the gripper low.

Coaching note: carry it down into the tote, then let go. These items are the ones that must not be dropped.

Violation: Card not put back

Visible cue: the location card is left out of its clip, the witness card is left off its spot, or either card is still in the gripper when the next step begins.

SOP rule broken: Steps 1.2 and 4.2, each card goes back to its own place before the next step starts.

Coaching note: the return is part of the step, not a gap between steps. Lower until it stands or lies, then let go.

Violation: Lid closed wrong

Visible cue: the lid is left standing part way, an edge or the item is caught under it, a gap shows along the seam, or the lid is pulled shut while the tote lifts off the table.

SOP rule broken: Step 5.1, the lid lies flat with no gap along the seam and nothing caught under it, and the tote stays on the table.

Coaching note: push it past upright and let it fall, then press the seam flat. A seal over a gap seals nothing.

Violation: Seal placed wrong

Visible cue: the seal body sits wholly on the lid or wholly on the front wall, runs along the seam instead of across it, misses the seam, or is put on with the seal number turned away from the camera.

SOP rule broken: Step 5.2, half of the seal body on the lid and half on the front wall, across the seam, with the number facing the camera.

Coaching note: across the seam, half and half, number facing out. A seal that does not bridge the seam is not a seal.

Violation: Seal pushed into place

Visible cue: the tag is slid, nudged or turned after it has touched the tote, instead of being left where it landed.

SOP rule broken: Step 5.2, once the tag is down it is not moved.

Coaching note: nothing on this station is pushed into line. Line it up in the air, then lower it once.

Violation: Press skipped or short

Visible cue: the seal body or the stub is let go and left with no closed gripper press, or a press is shorter than 2 seconds.

SOP rule broken: Steps 5.3 and 6.2, press each sticky part for 2 seconds from the middle out to each edge.

Coaching note: hold the full 2 seconds. A quick tap lets a seal lift later, and a seal that lifts is a broken chain.

Violation: Seal left lifting or torn

Visible cue: at the end of Step 6 a corner of the seal body stands off the tote, a crease or bubble is left under it, or the seal body has torn or come away with the stub and been left like that.

SOP rule broken: Steps 5.3 and 6.1, the seal body lies flat with no lifted corner, and it stays on the tote when the stub is peeled.

Coaching note: peel only the stub. Hold the seal body down with the eyes on it, and press any lifted corner once.

Violation: Bad tag used

Visible cue: a tag that is torn, folded onto its own sticky side, or doubled with a second tag is carried to the tote and put down on it.

SOP rule broken: Step 5.2, a bad tag goes to the reject spot and the next tag is used.

Coaching note: a bad tag is never used. It goes to the reject spot and a fresh one comes out.

Violation: Seal number not logged, or logged wrong

Visible cue: the episode ends with no stub on the log card, the stub goes in a box that is not the next open one, a box is left open while a later box is filled, a second stub goes in a filled box, or the stub is pressed with its number turned away or covered.

SOP rule broken: Step 6.2, one stub in the next open box, inside the lines, with its number in view.

Coaching note: fill from the front left box to the back right box in order. The number has to be readable from above or the log is worth nothing.

Violation: Numbers do not match

Visible cue: the number on the stub pressed to the log card is not the same as the number on the seal body left on the tote, or a stub from an earlier tag is used.

SOP rule broken: Step 7, the number on the seal body and the number on the stub are the same.

Coaching note: the stub belongs to the tag that is on the tote in front of it. Never log a number that is not on this tote.

Violation: Reject item or bad tag put in the wrong place

Visible cue: an item that would not read is put in the tote, back in the bin, or on the log card; a bad tag is left on the table, put on the tote, or put back on the tag spot; or a good item is put at the reject spot.

SOP rule broken: Steps 2.2, 4.1 and 5.2, items and tags that fail go to the reject spot and nowhere else.

Coaching note: anything that failed goes to the reject spot. Nothing that failed goes back into the flow.

Violation: Spill cleaned up

Visible cue: after an item, a card, a tag or a stub lands somewhere it should not, the gripper is used to sweep, push, wipe or pick it up.

SOP rule broken: Steps 1 to 6, a spill is left where it falls and is cleared in the reset.

Coaching note: cleanup moves are not part of the task. The spill is data, the cleanup is noise.

Violation: Bad pick, carry, or release

Visible cue: an object moves from one grip to another without being set down first, two objects are held at once, a grip is shifted or taken again while the object is off the table, an item or a card or the tote slides along the table with no daylight under it, an item hangs tilted because it was taken off square, or an object is turned over in the air.

SOP rule broken: Steps 1 to 6, one object at a time, lifted straight up off the surface before any sideways move, gripped square, and never passed between grips.

Coaching note: set it down to change the grip, and lift before the carry. An item that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit passed

Visible cue: an item, a card, the tote or the log card is shoved, nudged or slid into place instead of being lifted clear and placed again, or a failed check is tried again more often than its step allows.

SOP rule broken: Steps 1 to 6, a bad placement is fixed by lifting the object clear and placing it again, and each check allows only the fix it states.

Coaching note: nothing on this station is pushed into line. If the grip or the approach is forcing the same fix again and again, let review flag it and move on.

Violation: Carry over a working spot

Visible cue: a card, a tag or a stub is carried over the open tote on its way to somewhere else, a carry passes over the log card, or a carry crosses the far side of the table.

SOP rule broken: Workspace layout, nothing is carried over the open tote on the way to somewhere else and nothing is carried over the log card.

Coaching note: go around, not across. Anything dropped into an open tote goes in the record with the item.

Violation: Dropped, collided, or knocked over

Visible cue: under command, the gripper or a carried object hits the bin, the clip, the tote, the standing lid, the scanner or the log card and knocks, tips, slides or moves it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.

SOP rule broken: Steps 1 to 6, keep each object on a clear path and let go only after a settled placement.

Coaching note: watch the whole path, not just the target. A knocked tote breaks the seal record.

Violation: Work outside the camera frame

Visible cue: a pick, a scan try, a placement, the lid closing, the seal press or the stub press happens partly or wholly outside the environment camera frame, or the seal number is never in view.

SOP rule broken: Steps 1 to 6, every move happens inside the camera frame and every number is readable in it.

Coaching note: the work has drifted outside the framed area. Set the spots back to their setup places.

Violation: Wrong episode ending

Visible cue: the episode ends with the item still in the gripper or still out of the tote, the lid open, the tote unsealed, no stub on the log card, a card off its spot, short of home, or the gripper not fully open.

SOP rule broken: Step 7, check the tote, the numbers, the log card and the cards, then go home with the gripper open and stop recording.

Coaching note: check first. Going home is the last thing that happens before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Hardware fault: gripper failure, drift, controller caused collision, or motor error.

A spot sits outside comfortable reach, so the move stalls at a joint limit or has to fold in on itself to work the spot. Move the spot and run the Setup checklists again.

Scanner fault: no beep and no light on a clean, square, fully shown code after three scan tries, and the same failure on a test read of the spare test card. Check the scanner before the next episode.

Bad item label: one that is smeared, blank, torn or wet, so no reader can take it. Set the item aside before the next episode.

Bad card: a location card or witness card whose code is smeared or worn, or that bends when the gripper closes on it.

Bad clip: one that will not hold the location card upright, or that lifts the bin when the card is taken.

Bad tag from the batch: one that will not come off its backing, has lost its stick, or carries a blank or smeared number, or whose stub number does not match its body number. Replace the batch before the next episode.

Bad tote: one that slides or lifts under a correct press, whose lid will not fall shut on its own or will not lie flat, or that is cracked.

Bad log card: one whose boxes are torn, wet or already full.

Pick bin holding too few items to finish the log card, or a tag batch holding too few tags. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the location card from its clip
2. Show the location label to the scanner

3. Put the location card back in its clip
4. Take one item from the pick bin
5. Show the item label to the scanner
6. Set the item in the open tote
7. Take the witness card from its spot
8. Show the witness card to the scanner
9. Return the witness card to its spot
10. Close the tote lid
11. Peel one tamper tag from its backing
12. Place the tag across the lid seam
13. Press the seal flat
14. Peel the seal stub off the tag
15. Press the stub in the next open box on the log card
16. Go home and end the episode